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COS10025 Technology in an Indigenous Context Project unit

School of Science, Computing, and Engineering
Technologies

Seminar Week 3: Culture and application technology in the Vietnamese context

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- A 10x10 grid of dots, consisting of 10 rows and 10 columns of small black dots. The dots are arranged in a regular, repeating pattern across the entire page.

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Acknowledgement of Country

We honor and celebrate the rich tapestry of Vietnam's Indigenous peoples, encompassing 54 distinct ethnic groups, including the Kinh, Tay, Thai, Muong, Co Tu, Ede, Khmer, Hmong, Cham and many others, whose ancestral stewardship of this land spans millennia.

We pay our deepest respect to our ancestors past, present and future for preserving Vietnam's cultures, history, timeless traditions and deep spiritual connection to its mountains, rivers, seas and forests.

We recognize their enduring heritage, resilience, and invaluable contributions to the nation's diverse identity, committing to learn from and uplift their stories with reverence and gratitude.

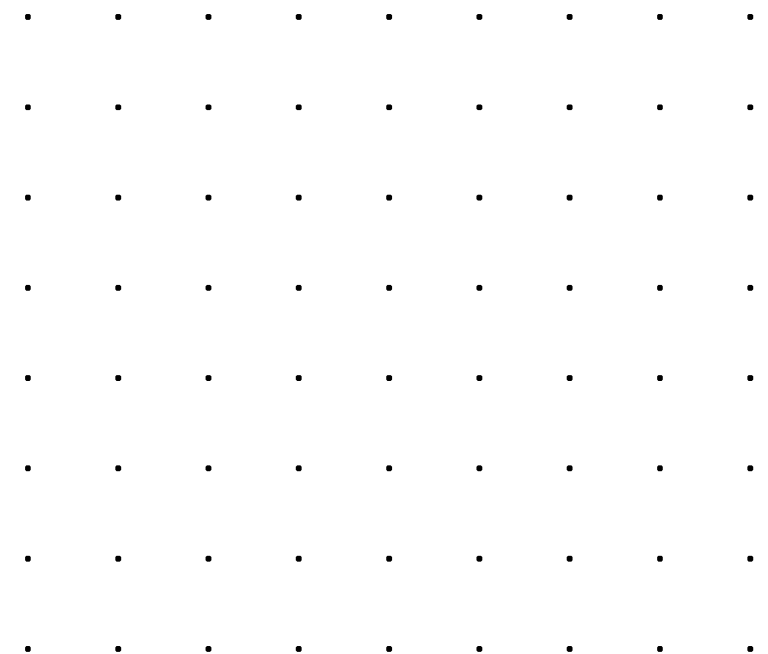


. . The worshipping of Hung Kings is a unique beauty in the spiritual culture of Vietnamese people

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Week 2 Recap



Designing with Country

Designing with Country is not possible without engaging with and be guided by Indigenous people.

Most importantly the knowledge held by Indigenous people should be an ongoing resource when designing with Country.

Design with Country is an important step to take –it allows us to gain a better understanding of how Indigenous culture can be respected and protected to allow culture to strengthen.

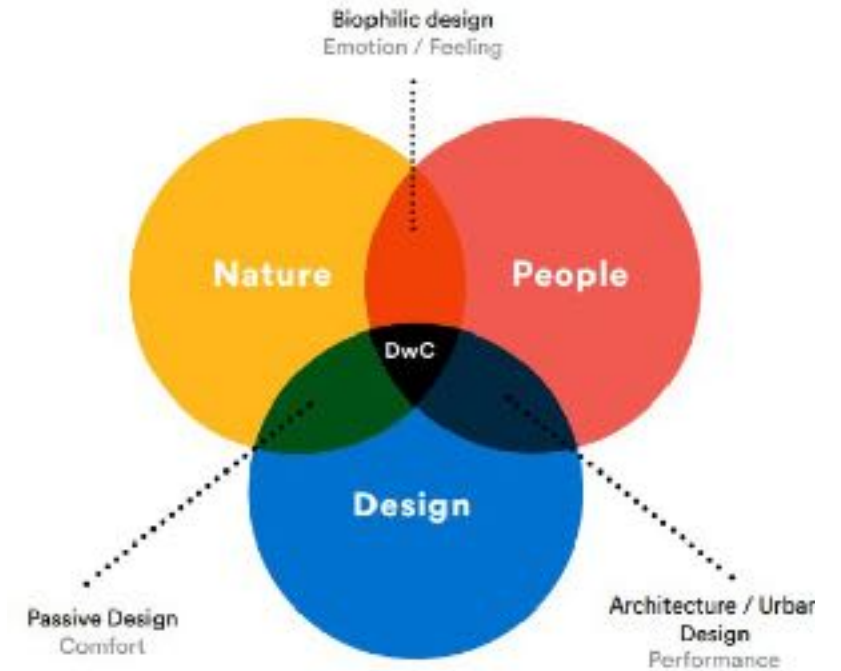
Three essential elements of design with Country –Nature, People, Design.

Three binary relationships –Architecture, Passive, Biophilic

Architecture considers design and people informed by nature

Passive considers design and nature, when utilized becomes environmental design

Biophilic design considers the essential connection between people and nature. When informed by design this specific relationship can be understood as an origin for Indigenous architecture



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Human-centred Vs Country-centred Design

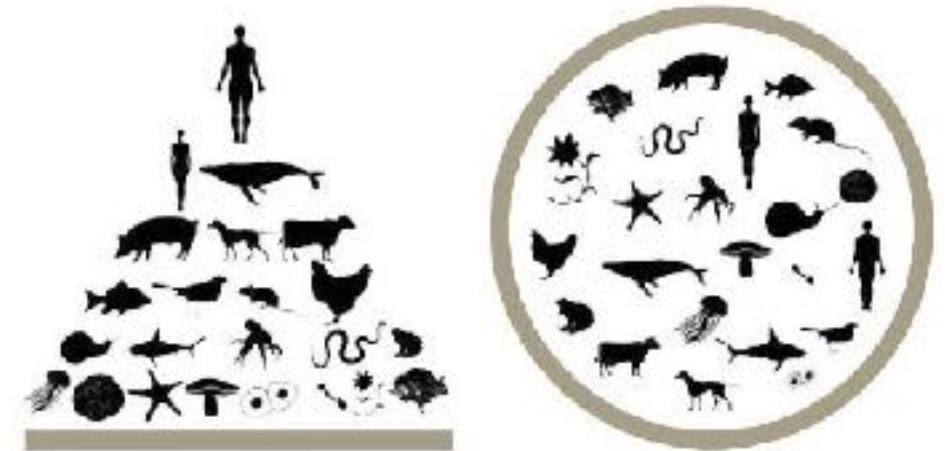
Sometimes referred to as Ego-centric or Eco-centric design

Indigenous people are referred to having an eco-centric world view, as the natural system involving people, plants and animals are considered when working with Country.

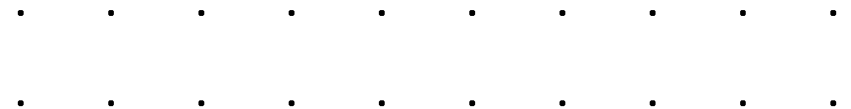
However, in contemporary design people and their needs are often prioritized over Country, where the landscape and nature are often reduced to second tier priorities.

By appreciating an Indigenous view of the world and incorporating people, resources, animals, waterways and plants equally we can make significant contributions and take further steps toward sustainable life.

When designing with Country, sustainability is not the aim but the outcome. By incorporating the needs of Country into the design phase of projects we can make these contributions to a sustainable future.



Ego-centric v Eco-centric diagram
adapted from Art Tawanghar, Designer, San Diego (2016)



Designing with Country

The following projects are some examples by Government Architecture NSW (GANSW)

These specific projects are able to demonstrate a genuine intent to build mutual respect between the Indigenous community and the design team. Demonstrating the meaning of Designing with Country.

-Casino Aboriginal Medical Centre –Casino NSW

-Wilcannia Health Service –Wilcannia NSW

-Kamay Botany Bay National Park –Sydney NSW



Photo: Toby Scott



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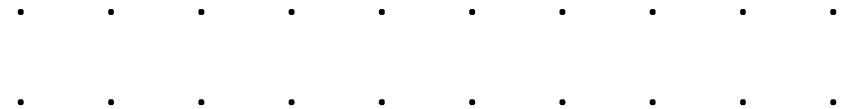


Culture and application technology in the Vietnamese context

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What will be covered today?

- Designing with Country projects
- 2 Way science
- Township challenges & opportunities



Na Pan primary school

Designing with Country examples

Location: Chieng Dong Commune, Yen Chau District, Son La, Vietnam

Completion Year: 2022

Project Area: 283 m²

Architects: 1+1>2 Architects

Material Use:

- New building constructed with adobe bricks, reminiscent of traditional pottery kilns.
- A circular multi-functional room built from river stones hand-collected by villagers.

Design Elements:

- Open design with numerous windows to maximize natural light and connect with the surrounding landscape.
- Window arrangement inspired by the traditional "Piêu scarf" dance of local people.

Community Involvement:

- Nearly 300 households contributed over 3,000 working days.
- Each person brought 30 pebbles from the river, totaling approximately 80,000 stones used in construction.



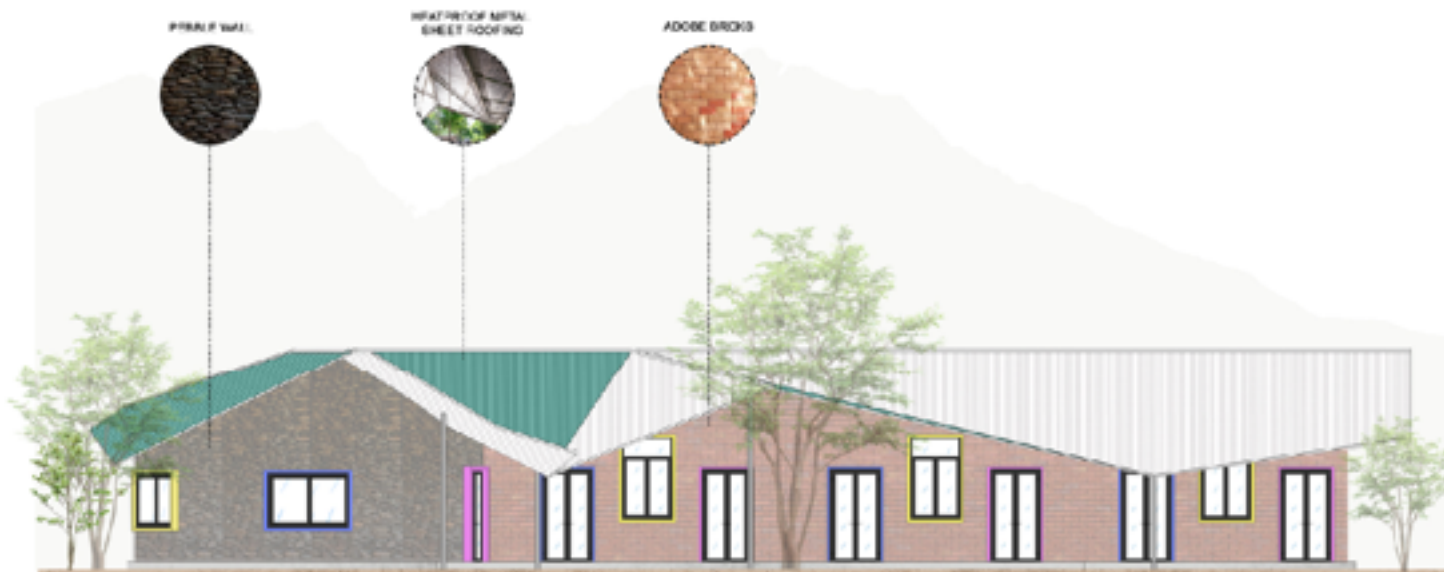


Na Pan primary school

Social & Cultural Impact

- The project not only transformed the school's appearance but also changed the mindset of the entire village.
- Through their contributions, villagers realized the importance of education, viewing it as essential as food and clothing.
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Na Pan primary school

Source: https://www.archdaily.com/1026959/na-pan-school-1-plus-1-2-architects?ad_source=search&ad_medium=projects_tab



VAC Library – Farming Architects



Location: Dương Nội, Hanoi, Vietnam

Completion Year: 2018

Project Area: 55 m²

Architects: Farming Architects

Concept & Design Philosophy

- **VAC System Inspiration:** The design is inspired by Vietnam's traditional VAC system—**Vườn (Garden), Ao (Pond), Chuồng (Cage)**—which integrates gardening, aquaculture, and animal husbandry to maximize resource efficiency. [Arquitectura Viva+1Mooool+1](#)
- **Urban Adaptation:** The project reimagines the VAC model for urban settings, creating a multifunctional space that combines a library, play area, and ecological education center.

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Architectural Features

- **Modular Wooden Framework:** The structure comprises a grid of wooden frames that support various functions, including bookshelves, seating, planters, and climbing spaces. [Arquitectura Viva](#)
- **Integrated Ecosystems:**
 - **Aquaponics Pond:** A fish pond that supports plant growth through nutrient-rich water.
 - **Vegetation:** Climbing plants and vegetables grow throughout the structure, providing shade and educational opportunities.
 - **Chicken Coop:** Incorporated into the design to demonstrate animal husbandry practices.
- **Sustainable Design:** The open framework allows for natural ventilation and lighting, reducing energy consumption.

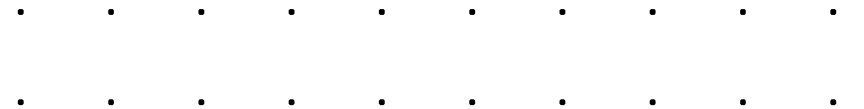
Educational & Community Impact

- **Children's Learning Space:** Designed as an interactive environment where children can learn about ecology, sustainability, and traditional farming practices.
- **Community Engagement:** Serves as a communal area that fosters interaction among residents, promoting awareness of sustainable living.

Source: https://www.archdaily.com/908873/vac-library-farming-architects?ad_source=search&ad_medium=projects_tab

Key factors to working with Country

- Deep Listening
- Indigenous led
- Community/Place specific
- Shared benefit
- Respecting knowledge and place



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2-Way Science

Ngaparrtji Ngaparrtji

Two-way

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2 way Initiative

This book is titled 'Ngaparrtji Ngaparrtji Twoway'.

Ngaparrtji Ngaparrtji is a phrase used by several Western Desert Aboriginal languages that loosely translates as 'I give you something, you give me something'. It is a complex cultural concept of reciprocity and exchange.

Ngaparrtji Ngaparrtji is used here to elaborate on the term 'two-way' which underpins the Two-way Science Initiative and the principles that inform its development and implementation. It speaks to the obligation of education systems to ensure Aboriginal knowledges are included alongside other knowledges in the curriculum.

It also speaks of the obligation of schools to value and respect the gift of this knowledge and ensure that Aboriginal people are acknowledged for their participation in the development of local culturally responsive teaching and learning programs. It refers to the complex interplay between Aboriginal scientific understandings of Country and culture, and Western scientific knowledge.

Aboriginal people are the first scientists, the first engineers and the first educators. This work is grounded in the critical importance of ensuring more than 60,000 years of wisdom and knowledge of Aboriginal people has its rightful place in our education systems

Source: <https://static1.squarespace.com/static/5fd6a04f674bc6778f41eb42/t/604ea8effb956624c62d0dca/1615767899729/2+way.pdf>



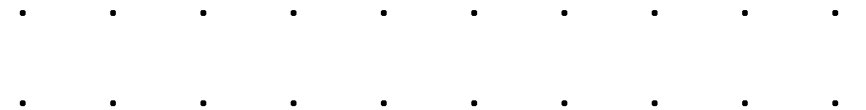
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2-Way Science Principles

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Reciprocity

The efforts of Aboriginal communities must be recognised. That can be through remuneration, advocacy and support for local initiatives.



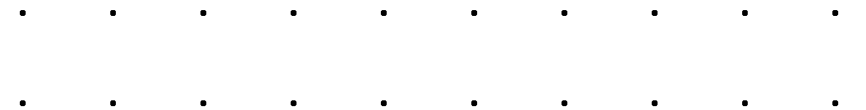
Co-design

Two-way Science strategies, procedures, schedules, monitoring and evaluation, professional learning and education resources must be developed through codesign with Aboriginal people and communities

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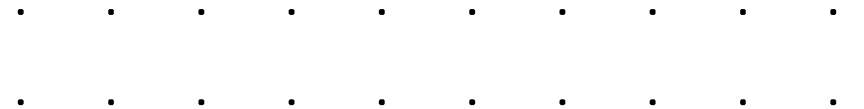
Listen to Country

Time must be spent on Country, listening to Country and learning from Country



Local place-based solutions

Aboriginal cultures, languages and communities are different from one another and require local placebased programs to support the learning needs of students. Culturally responsive approaches to place and space must include traditional owners and build on the strength of diversity of Aboriginal cultural identities of students and families connected to the school.



Value and respect what people are already doing

Aboriginal communities and schools have a long history of developing innovations in education delivery. These innovations and the people that developed them should be respected and supported to action them within the scope of the project.

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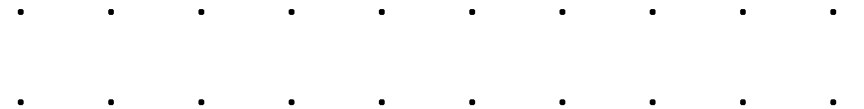
Cultural safety

The project systems and structures must enable the participation of Aboriginal people in ways that make people feel welcome and safe, and recognise that this will be different in different contexts and for every individual. This includes, for example, the way engagement and consultation takes place, where and how workshops are held, knowledge sharing protocols and the impact of colonisation on language and cultures.

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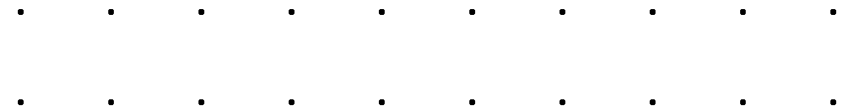
Aboriginal intellectual and cultural property

Aboriginal intellectual and cultural property is owned by the authors and traditional owners of that property. Consent and permission must be obtained whenever intellectual and cultural property is used by anyone for any purpose.



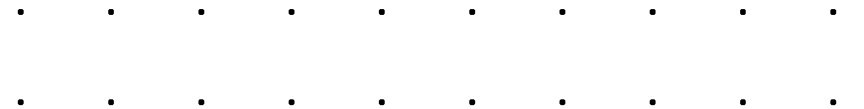
Co-authorship

Projects for Aboriginal people must be developed by, or with, Aboriginal people. Co-authorship must be acknowledged where Aboriginal people have had an input into the development of projects as well as any written or audio visual material.



Commitment and sustainability

Support and grow Aboriginal-led initiatives to produce long term sustainable outcomes for schools and communities



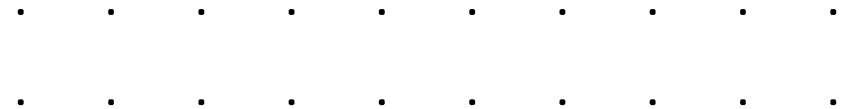
Work at the pace of Aboriginal communities

Aboriginal people must be given the opportunity to build projects according to local cultural protocols and practices with the full input of communities and Elders who have diverse commitments and geographic locations.



Take time to build relationships and trust

newcomers to communities must take the time to listen and build relationships and trust during project development



Define success

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Two-way science initiative

Two-way Science provides an opportunity for all students to learn about science from the world's oldest continuing cultures.

The Two-way Science Initiative aims to:

- build respect for Aboriginal histories, cultures, people, knowledges and experiences amongst school staff and students
- strengthen Aboriginal student wellbeing, engagement and achievement.

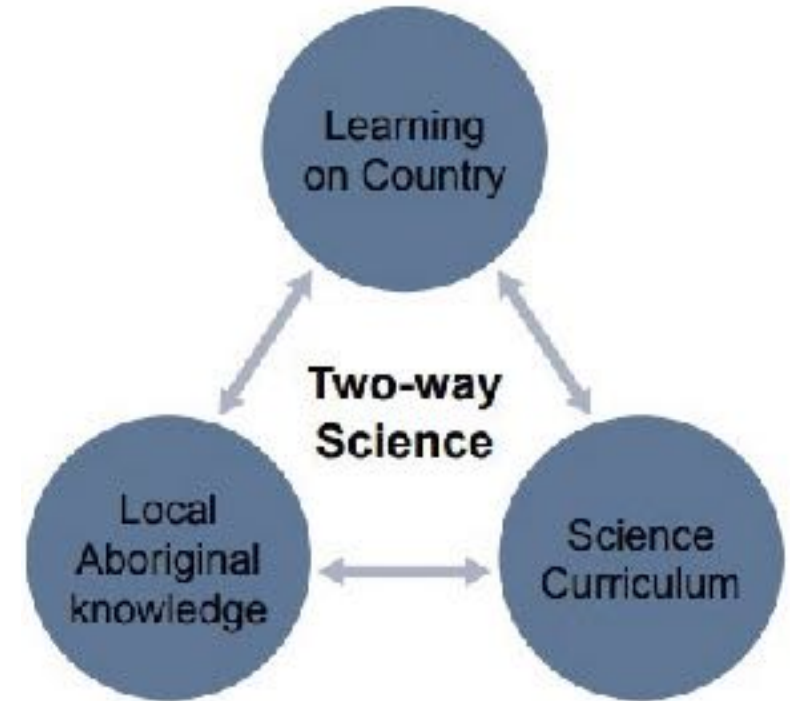


Two-way approach

Two-way Science privileges local Aboriginal knowledge, empowerment and leadership but asks all staff to work and learn together.

A Two-way Science program contains three essential elements:

- Aboriginal knowledge of the local community
- Learning on Country led by local Aboriginal experts
- The science curriculum



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Two-way approach case study

WASI - Application of AI Technology in Increasing Agricultural Productivity in Daklak, Vietnam



What are 2-way science strategies in this project?

Enfarm introduced a **smart device** for **nutrient measurement** in soil and plants, helping farmers:
Reduce fertilizer use by providing precise recommendations.
Increase agricultural productivity.

Cut farming costs and minimize environmental impacts.

The device uses **AI** to **analyze soil samples** quickly and accurately, achieving laboratory-standard precision.

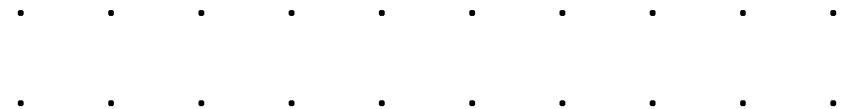
Enfarm's AI product was ranked **Top 5 in the AI Awards 2024** and recognized among **Vietnam's Best Products 2024**.

Videos

<https://www.youtube.com/watch?v=BbtReu9I9vQ>

<https://www.youtube.com/watch?v=N9QHa8k0rwY>

<https://enfarm.com/cong-nghe-enfarm-lap-dat-tai-vuon-sau-rieng-co-thu-huyen-krong-pac>



What are 2-way science strategies in this project?

The AI systems in this project not only introduce technology in a top-down manner but also **integrate traditional farming knowledge** (for example: seasonal cycles, soil management practices, and weather forecasting based on natural signs) from ethnic groups such as the Ê Đê and M'Nông.

The goal is to **adapt technology to fit local farming customs**, rather than imposing industrialized methods.

In fact, during **farmer training sessions**, local farmers' experiences are documented and incorporated into the AI algorithms to improve their effectiveness.



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Full-size map at: <https://hocvientrithuc.com/ban-do-phan-bo-cac-dan-toc-viet-nam/?>

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Challenge: Businesses Struggle to Cope with Transportation Difficulties from the Vietnam Logistics Business Association (VLA)

Widespread Transportation Challenges

- **Skyrocketing Freight Rates:** Exporters are facing unprecedented increases in shipping costs.
- **Container Shortages:** A significant lack of empty containers for loading goods has been reported.
- **Limited Vessel Space:** There's a scarcity of available slots on ships, especially due to a surge in exports in late 2020 and early 2021.
- **Overloaded Inland Container Depots (ICDs):** Key ICDs are experiencing congestion, hindering the timely drop-off of export containers.
- **Delayed Empty Container Rotation:** The movement of empty containers between ports, ICDs, depots, and factories has slowed, exacerbating logistical issues.

Infrastructure Bottlenecks

- **Heavy Reliance on ICDs:** Approximately 87% of exports at Cái Mép – Thị Vải (CM-TV) port are routed through ICDs and river ports before being transported via barges.
- **Overcapacity at ICDs and River Ports:** The rapid increase in cargo volume has led to these facilities operating beyond their capacity, posing risks to supply chain efficiency.
- **Global Supply Chain Disruptions:** Similar bottlenecks are occurring worldwide, with Vietnam also experiencing challenges, though not yet at the level of complete port gridlocks.



Proposed Solutions?

Reference: <https://vla.com.vn/doanh-nghiep-xoay-xo-voi-kho-khan-ve-van-chuyen/>

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Challenge: Current Situation of Healthcare Human Resources in Cao Bằng

Population and Geographic Features:

- Cao Bằng Province has a population of over 530,000 people, with eight major ethnic groups.
- Ethnic composition: Tày (40.97%), Nùng (31.08%), Dao (10.08%), and Kinh (5.76%).
- The province's mountainous terrain and predominantly ethnic minority population pose challenges for healthcare delivery.

Shortage of Qualified Medical Staff:

- The provincial healthcare sector employs 3,226 officials and public employees.
- Education levels:
 - Specialists II, Master's degrees, and Specialists I account for only 12.3%.
 - University graduates: 28.2%.
 - College graduates: 21.8%.
 - Intermediate (secondary) level: 29.1%.
- The healthcare workforce is both quantitatively insufficient and professionally underqualified.



Project Opportunities?

Reference: <https://baocaobang.vn/-45308.html?>

Challenge: Severe Water Shortage in Lục Khu, Cao Bằng

Impact on Residents:

- Local communities have long struggled with limited access to clean water for daily use.
- The scarcity affects various aspects of life, including hygiene, agriculture, and overall well-being.

Community Efforts and Aspirations:

- Despite the hardships, residents remain hopeful for sustainable solutions to their water needs.
- There is a strong desire for infrastructure projects that can provide reliable water sources.

Call to Action:

- The situation underscores the need for targeted interventions from authorities and organizations to address the water crisis in Lục Khu



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Current projects:

Construction of Water Reservoirs on Limestone Mountains

Source: <https://sokhcn.caobang.gov.vn/tom-tat-dt-da-20062010-64960/ung-dung-cong-nghe-xay-dung-mo-hinh-ho-chua-nuoc-tren-nui-da-voi-556381>

Urban Water Supply Projects

Source: <https://ns.mard.gov.vn/tin-trong-nuoc/cao-bang-tang-ty-le-dan-cu-duoc-su-dung-nuoc-sach-con-nhieu-kho-khan>



A water monitoring station in Xuan Dai Bay in Vietnam

Source: <https://www.uts.edu.au/case-studies/water-monitoring-solution-vietnamese-aquaculture>

Water Supply Projects in Rural Areas

Source: <https://baocaobang.vn/can-co-phuong-an-giai-quyet-cac-cong-trinh-cap-nuoc-sach-nong-thon-tap-trung-kem-hieu-qua-3172772.html>

Groundwater Exploration and Protection

Source: <https://nongnghiepmoitruong.vn/cao-bang-trien-khai-nhieu-bien-phap-bao-ve-nguon-tai-nguyen-nuoc-d339750.html>

Project Opportunities?

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